

WHICH TESTS CATCH FALSE PLANETS IN ARCHIVAL RADIAL VELOCITIES? AN INJECTION-CALIBRATED VETTING LADDER TESTED ON PUBLISHED SIGNALS WITH KNOWN FATES

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ABSTRACT

Many low-mass planets claimed from archival radial velocities have been withdrawn as stellar activity, aliases or noise. We measure how well common vetting tests separate planets from impostors on the real sampling and noise of HARPS-RVBank. We implement a ladder of interpretable tests (false-alarm probability, aliases, phase and amplitude stationarity, activity indicators, rotation harmonics) in tested open-source software and run it blind on 16 000 simulated series from 260 HARPS stars, with planets and evolving quasi-periodic activity injected into permuted real noise. The false-alarm probability depends on how the jitter is fitted: held at its planet-free value it is conservative by a factor of 4–8, and rescaled with the signal it is liberal by 1.8 at 1%. Many activity impostors fall inside the broad lines of evolving rotation, not on its exact harmonics, more so for short-lived spots. No single test is decisive (best area under the ROC curve 0.78, detection statistics 0.74); all rungs together reach 0.917, with indicator power the most important input. Of 17 published signals later refuted, 11 are not significant in a blind search of the full archive; a rule with thresholds fixed before this benchmark (its features were revised afterwards) rejects 5 of the other 6 and 2 of 21 significant upheld signals, counts that shift with the window around the claimed period. On the archive as of January 2022, the ladder ranks signals later published as planets near the top (6 of the top 10, 5 on TESS candidate hosts; 1.0 expected), though not measurably better than the false-alarm probability; one of two predictions tested with newer spectra failed. We also report the 84 defects found in the AI-assisted analysis behind this work, coded by two independent instances of the model: overclaims were the largest category, few were numbers miscopied from code, and an independent reviewer found most.

Subject headings: techniques: radial velocities – methods: data analysis – methods: statistical – planets and satellites: detection – stars: activity

1. INTRODUCTION

Public radial-velocity (RV) archives now cover two decades. The HARPS-RVBank (Trifonov et al. 2020; Perdelwitz et al. 2024) gives uniformly re-derived velocities and activity indicators for more than 5000 stars observed with HARPS (Mayor et al. 2003), and it has already been searched for planets (Sreenivas et al. 2022). Low-amplitude signals in such data have a mixed record. Signals later withdrawn turned out to be produced by the sampling and the noise (Rajpaul et al. 2016), by stellar rotation or its harmonics (Robertson et al. 2014; Santos et al. 2014; Robertson et al. 2015; Carleo et al. 2020; Burrows et al. 2024), or by an alias of rotation that did not hold phase when new data arrived (Lubin et al. 2021). The tests used to screen candidates are well known individually: false-alarm probabilities (Baluev 2008), including against red noise (Ejaz et al. 2026), alias comparisons, activity-indicator periodograms, signal growth and stacked periodograms (Mortier & Collier Cameron 2017), apodised sinusoids (Hara et al. 2022a) and false-inclusion probabilities (Hara et al. 2022b). What is rarely measured is how well each of them separates planets from impostors on the sampling and noise of real archival series, and how they perform on signals whose fate is known.

Blind challenges have measured how well activity models

recover planets in simulated data sets (Dumusque et al. 2017; Zhao et al. 2022), and a neural network trained on simulations has been proposed to replace the significance test and classify signals as planetary or not (Nieto & Díaz 2023). Here we ask a complementary question for archival searches: given a significant periodic signal in a long, irregular, single-instrument series, which tests tell a planet from an impostor, and how often do they fail? We answer it in four steps.

1. We implement a ladder of interpretable tests as tested, open software (Sect. 3).
2. We calibrate every test on simulations built from the real sampling and noise of 260 HARPS-RVBank stars, with planets and evolving stellar activity injected and every significant peak labelled (Sects. 4 and 5).
3. We apply the ladder, a rule whose thresholds were fixed before the benchmark was run, and classifiers trained only on the simulations to published signals around HARPS-RVBank stars whose later verdict is known (Sect. 6), and to the archive as it stood in January 2022, against the planets published since (Sect. 6.5).
4. Because the analysis that motivated this work was written with an AI assistant, we report the defects that checks of it found, coded by two independent instances

of the model, as a case study in verifying AI-assisted analysis (Sect. 8).

The candidate that motivated the work, HD 297396 b (Fraser, in preparation), is the worked example of Sect. 7.

2. DATA

We use the HARPS-RVBank table of [Perdelwitz et al. \(2024\)](#) (CDS J/A+A/683/A125, the version of 2026 April 20), with 252 615 spectra of 5239 stars taken before 2022. For each spectrum it gives the SERVAL velocity ([Zechmeister et al. 2018](#)) corrected for the nightly zero point, its uncertainty, and eight activity indicators: the differential line width (dLW), $H\alpha$, the NaI D1 index, the chromatic index (CRX), the full width at half maximum (FWHM), contrast and bisector span of the cross-correlation function, and R'_{HK} .

The same instrumental cuts, none of which uses model residuals, apply to every star: RVBank flag zero, signal-to-noise ratio at least 10 in order 55, absolute drift at most 3 m s^{-1} , and a finite uncertainty below 20 m s^{-1} (91% of all spectra pass). Spectra are averaged by night with inverse-variance weights, separately before and after the 2015 fibre upgrade ([Lo Curto et al. 2015](#)); a night runs from local noon to local noon at La Silla (16:43 UT). The two eras receive independent offsets and jitters. For every star, simulated or real, nightly epochs more than 15 robust standard deviations from a per-era quadratic are dropped, which removes misidentified or grossly deviant spectra (8 epochs in 3 benchmark hosts) but keeps merely discrepant nights, and a quadratic trend is fitted jointly with everything else. We write N for the number of nightly epochs, T for the baseline and $n_{\text{era}} \leq 2$ for the number of eras.

3. THE VETTING LADDER

3.1. Periodogram and detection

For velocities y_i with uncertainties ϵ_i and era labels $g(i)$, the model has per-era offsets and per-era jitters s_g , so that the variance of epoch i is $\sigma_i^2 = \epsilon_i^2 + s_{g(i)}^2$. With the jitters fixed, the improvement in χ^2 from adding a sinusoid at frequency f is

$$\Delta\chi^2(f) = \chi_{\text{offsets}}^2 - \chi_{\text{offsets}+a\cos 2\pi ft+b\sin 2\pi ft}^2, \quad \chi^2 = \sum_i r_i^2/\sigma_i^2. \quad (1)$$

Offsets are removed by projecting the whitened sinusoids and data (each divided by σ_i) onto the complement of the whitened era indicators, so each frequency costs $O(Nn_{\text{era}})$ operations and many simulated series are processed in one matrix product. The search band is 1.2–500 d on a grid of three frequencies per $1/T$.

The jitters matter. We first fit them by maximum likelihood with offsets only, and take the candidate frequency f_0 as the highest peak of $\Delta\chi^2(f)$ with those jitters (or, for a claimed period, the highest point within $\pm 0.5/T$ of it, half a resolution element on either side; Sect. 6.4 gives the effect of this choice). A real signal inflates the planet-free jitters and so deflates its own $\Delta\chi^2$. We therefore rescale all jitters by one common factor fitted with the sinusoid at f_0 in the model, and use as detection statistic the likelihood ratio $z_0 = 2(\ln L_1 - \ln L_0)$, with the jitters at their planet-free values under L_0 and rescaled under L_1 . (Refitting each era's jitter separately under L_1 can collapse the jitter of an era with few epochs onto a sinusoid that happens to fit them; one common factor cannot.) All later rungs use the rescaled jitters. The false-alarm probability (FAP) of

z_0 over the band is the approximation of [Baluev \(2008\)](#) for a sinusoid in noise of known variance,

$$\text{FAP} \approx 1 - (1 - e^{-z_0/2})e^{-\tau}, \quad \tau = W e^{-z_0/2} \sqrt{z_0/2}, \quad (2)$$

with $W = (f_{\text{max}} - f_{\text{min}}) \sqrt{4\pi \text{Var}_w(t)}$ and $\text{Var}_w(t)$ the variance of the epochs weighted by $1/\sigma_i^2$. It bounds the FAP from above as the FAP tends to zero; how it performs on real noise with a fitted jitter is measured in Sect. 5.1. For a period known in advance, the single-frequency probability is $e^{-z_0/2}$.

3.2. The rungs

Given f_0 , the ladder computes the features of Table 1.

Amplitude. The semi-amplitude K and its uncertainty from the sinusoid fit, K/σ_K , and K over the rms of the velocities about the offsets.

Period. The candidate aliases of f_0 are its first-order aliases through the sidereal day, the synodic month and the year and through the three strongest peaks of the star's own spectral window. Those inside the search band are refined within $\pm 0.5/T$ (inside the band), and any that refines to within $1.5/T$ of f_0 is dropped, since it has found the candidate's own peak. The feature is the margin in $\Delta\chi^2$ between f_0 and the strongest remaining alias.

Stationarity. A Keplerian keeps amplitude and phase; rotational modulation does not. We fit the sinusoid at f_0 , with a common phase origin, separately to the two halves of the series and to four blocks with equal numbers of epochs. Each fit gives (a, b) with a covariance matrix. About their generalised-least-squares mean μ , the deviations are split into the component along μ (amplitude) and across it (phase), and each sum of squared standardised deviations is compared with a χ^2 distribution with $n_b - 1$ degrees of freedom, n_b being the number of sub-samples. For four blocks this gives the probabilities `block_phase_p` and `block_amp_p`, and the rms phase deviation; for the halves, the phase and amplitude differences in units of their uncertainty and the imbalance between the halves' share of $\Delta\chi^2$ and their share of epochs. Because f_0 is fitted to the whole series, any constant phase drift between the halves is absorbed into it, so the half-phase test is weak by construction; the block test, with four sub-samples, retains power, and its calibration on simulated planets is given in Sect. 5.4. We also follow $\Delta\chi^2$ as epochs accumulate in time order and report its Spearman correlation with the number of epochs (cf. [Mortier & Collier Cameron 2017](#); [Sreenivas et al. 2022](#)), and we replace the constant amplitude by a Gaussian envelope over a grid of centres and widths (0.1–0.5 of the baseline) and report the gain in $\Delta\chi^2$ (cf. the apodised sinusoids of [Hara et al. 2022a](#)).

Activity. Each indicator is fitted with per-era offsets and the same quadratic trend as the velocities, which are removed, and is divided by its robust scatter over the whole series. Values more than 15 robust standard deviations from the fit, the level used for the velocities, are dropped and the fit repeated, since a least-squares trend tilted by one gross outlier leaves a spurious long-period signal. Indicators with values on fewer than 60% of the epochs are dropped. Indicators carry no usable errors, so each era receives the closed-form maximum-likelihood jitter (the era's standard deviation). The feature `ind_max_dchi2` is the largest $\Delta\chi^2$ of any indicator within $\pm 0.5/T$ of f_0 (eleven frequencies); with no usable indicator it is zero. We also report the largest absolute correlation of an indicator with the velocities. As a rotation proxy we take, among dLW, $H\alpha$, FWHM and R'_{HK} , the most significant local

TABLE 1
LADDER FEATURES AND THE QUESTION EACH ANSWERS.

Feature	Definition
dchi2, log10_fap	likelihood ratio z_0 ; its Baluev FAP over 1.2–500 d
snr_K, K_over_rms	K/σ_K ; K over the rms about the offsets
alias_margin	$\Delta\chi^2(f_0)$ minus the largest alias $\Delta\chi^2$ inside the band
half_phase_z, half_amp_z	phase, amplitude difference between halves (in σ)
half_imbalance	share of $\Delta\chi^2$ – share of epochs in the first half
block_phase_p, block_amp_p, block_phase_rms	constancy of phase and amplitude in four blocks; rms phase deviation
growth_rho	Spearman ρ of $\Delta\chi^2(f_0)$ with epochs in time order
apod_gain	$\Delta\chi^2$ gain of a Gaussian envelope
ind_max_dchi2, ind_max_absr	largest indicator $\Delta\chi^2$ near f_0 ; largest $ r $ with the RVs
act_log10_fap	FAP of the rotation proxy (2.5–200 d, seasonal aliases excluded)
harm_dist, harm_alias_dist, harm_missing	distance to kf_{proxy} and to its aliases ($1/T$ units); no proxy
log_n, log_ncyc	$\log N$; $\log(Tf_0)$. Reported, not used by the classifiers

maximum over 2.5–200 d, excluding frequencies within $1/T$ of one and two cycles per year, where seasonal sampling puts spurious power; its Baluev FAP is multiplied by the number of indicators searched, and we write f_{proxy} for its frequency. When that FAP is below 1%, **harm_dist** is the distance of f_0 from the nearest of kf_{proxy} , $k = 1-4$, in units of $1/T$, and **harm_alias_dist** the distance from the nearest first-order alias of one of them; otherwise both are undefined and the classifiers receive a flag, **harm_missing**.

Known signals. When a system has accepted planets, they are removed first by Keplerian fits with fixed periods. For fixed eccentricity and time of periastron the orbit is linear in $K \cos \omega$ and $K \sin \omega$, so the orbital elements are found by a grid in (e, t_p) and simplex refinement, by backfitting over the planets, jointly with the offsets and the quadratic trend.

3.3. An a-priori rule

Before any benchmark signal was run through the pipeline we wrote down a rule-based verdict (released with the code): a candidate passes if its FAP is below 1%, it beats its aliases, its phase is constant across blocks ($p > 0.01$) and halves ($< 3\sigma$), no indicator has $\Delta\chi^2 \geq 13.8$ near f_0 , and it is not within one resolution element of a rotation harmonic. A rung that cannot be computed does not reject: with fewer than three usable blocks or no halves the phase rung passes, and without a significant rotation proxy the rotation rung passes. The thresholds came from nominal significance levels and from the analysis of HD 297396 b, so that candidate’s passing the rule (Sect. 7) is partly circular. They are not nominal error rates: 13.8 is the single-frequency $p = 10^{-3}$ point of a χ^2_2 distribution, but it is applied to the maximum over eight indicators and eleven frequencies. We therefore report what each rung does to simulated planets (Sect. 5.4). The rule does not use **harm_alias_dist**, which we added after the first benchmark pass; results with it are labelled as such. The thresholds have not changed since they were frozen, but the features they apply to have (Sect. 6.4).

4. SIMULATIONS ON REAL SAMPLING AND NOISE

We selected 260 stars at random from those with 40–250 nightly epochs, a baseline of at least 1500 d, a median un-

certainty of at most 3 m s^{-1} , dLW and H α on at least 80% of the spectra, and an RV standard deviation of at most 20 m s^{-1} (which excludes binaries and giant planets), and excluded every benchmark host and HD 297396.

Noise. Each simulated series starts from the star’s real nightly velocities and indicators with one random permutation within each era, shared by the velocities, the indicators and the uncertainties. The sampling, the errors and the joint distribution of velocity and indicator noise, including its tails, are real; any coherent signal the star had, and any noise correlated between nights, is destroyed. The star’s series is clipped for outliers before the permutation, and each simulated series is fitted with the same quadratic trend as a real star.

Planets. Keplerians with P log-uniform in 1.2–300 d, eccentricity from the distribution of [Kipping \(2013\)](#) redrawn above 0.8, random orientation and phase, and K log-uniform in 0.15–3 times the scatter σ of the noise series. We describe each case by its expected K/σ_K , with $\sigma_K = \sigma\sqrt{2/N}$.

Activity. We use the FF’ picture ([Aigrain et al. 2012](#); [Rajpaul et al. 2015](#)): a latent process $G(t)$ with a quasi-periodic kernel $k(\tau) = \exp[-\tau^2/2\lambda^2 - \sin^2(\pi\tau/P_{\text{rot}})/2w^2]$ and its derivative $G'(t)$, drawn jointly at the epochs from the analytic covariance of (G, G') . The velocities receive $A[\sqrt{1-\varphi^2}G + \varphi G']$, φ being the share of the rotational term G' , with P_{rot} log-uniform in 3–150 d, λ log-uniform in 1–30 rotations, w uniform in 0.25–1.2, A log-uniform in 0.2–3 times σ and φ uniform in 0–0.9. Both K and A scale with σ and not with N , so the number of epochs carries no information about the class. How strongly activity shows in the indicators is the most consequential assumption of the simulations, so we vary it: in four cases out of five each indicator j receives $A_j[\cos \theta_j G + \sin \theta_j G']$ times its own scatter, with a common amplitude log-uniform in 0.1–3, a factor $10^{N(0,0.3)}$ for each indicator and a phase lag θ_j uniform within $\pm 45^\circ$ of 0 (all indicators except the bisector) or of 90° (bisector); in the fifth case the indicators receive nothing. The indicator amplitudes are drawn independently of the RV amplitude A .

Cases and labels. Each case is a planet alone (35%), a planet with activity (25%), activity alone (30%) or noise alone (10%). The ladder is run blind on the highest peak. That peak is labelled, in this order of precedence: a *planet* if it lies within $1/T$ of the injected orbital frequency f_p ; a *planet alias* if within $1/T$ of $2f_p$, $3f_p$, $f_p/2$ or a first-order alias of f_p ; a *rotation harmonic* if within $1/T$ of kf_{rot} ($k = 1-4$, with $f_{\text{rot}} = 1/P_{\text{rot}}$ the injected rotation frequency) or a first-order alias of one; a *rotation line* if within $\max[1/T, 1/(\pi\lambda)]$ of kf_{rot} , $k = 0-4$, that is inside the $\pm 2\sigma$ width of a line of the quasi-periodic power spectrum, whose lines are Gaussians of standard deviation $1/(2\pi\lambda)$ (for $k = 0$, inside the low-frequency envelope of the evolving amplitude); otherwise *other activity* if the series has activity, and *noise* if not. Every case is reproducible from its seed.

5. RESULTS ON SIMULATIONS

The simulation set has 16 000 cases on 260 stars with a median of 73 nightly epochs: 5576 planets alone, 4035 planets with activity, 4837 activity alone and 1552 noise alone.

5.1. False alarms

For the false-alarm calibration we drew 40 noise-only series per star (10 400 in all) of two kinds: the star’s own velocities permuted within each era, and Gaussian noise with the star’s error bars and fitted jitters. Each went through the same clip,

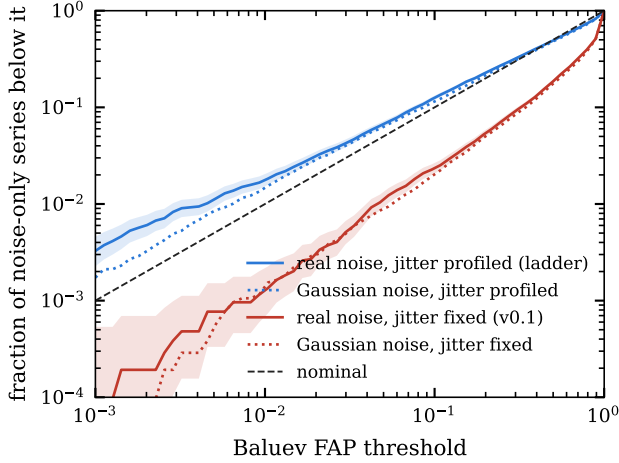


FIG. 1.— False-alarm calibration: the fraction of noise-only series whose highest peak has a Baluev FAP below a given threshold, for the likelihood-ratio statistic of the ladder (blue) and for $\Delta\chi^2$ with the jitter fixed at its planet-free value (red), on the stars’ own permuted velocities (solid, with 95% intervals) and on Gaussian noise (dotted). The dashed line is the nominal rate.

trend and detection step as the simulations (Fig. 1). Neither common treatment of the jitter gives nominal rates. With the jitter held at its planet-free value, the statistic of our first version, the Baluev FAP of the highest peak falls below 1% in only 0.12% of the real-noise series (95% interval 0.07–0.21%) and below 10% in 2.3%: conservative by a factor of 4–8 between 1 and 10%, because a peak’s own power inflates the jitter that weights it. With the jitter rescaled under the signal model, as in the ladder, the approximation errs the other way: 1.8% of real-noise series fall below 1% (1.5–2.0%), 13% below 10% and 0.33% below 0.1%, a factor of 1.8 at 1% and 3.3 at 0.1%. For Gaussian noise the factor at 1% is 1.5, so part of the excess comes from fitting the jitter scale, which the known-variance approximation does not count, and part from the heavy tails of real velocities. On these stars a Baluev FAP of 0.44% (95% interval 0.29–0.54%) corresponds to a real false-alarm rate of 1%, and 0.043% (0.022–0.065%) to 0.1%. We keep the frozen 1% threshold in the rule and report its real rate. Permutation makes the noise white between nights, so none of these rates includes correlated noise on timescales longer than a night; false-alarm levels for red noise (Ejaz et al. 2026) address that part.

5.2. Completeness

Planets alone are recovered at the right period with FAP below 1% in half the cases at an expected K/σ_K of 6.4 and in 90% at 11 (Fig. 2), with 93–96% recovery across five period ranges between 1.2 and 300 d at expected $K/\sigma_K > 10$. Of the 2124 series with a planet alone and a significant highest peak, 10.4% (220) peak at the wrong period: 100 at an alias or harmonic of the planet and the rest on a noise peak that outgrew a weak planet.

5.3. What the impostors are

Of the 4931 cases with a significant highest peak, 2871 (58%) are the injected planet and 2060 are impostors: 671 at a rotation harmonic or its alias, 629 inside a line of the activity spectrum but more than one resolution element from a harmonic (189 of them in the low-frequency envelope), 436 elsewhere in series with activity, 176 at an alias or harmonic of the planet, and 148 from noise. These proportions follow

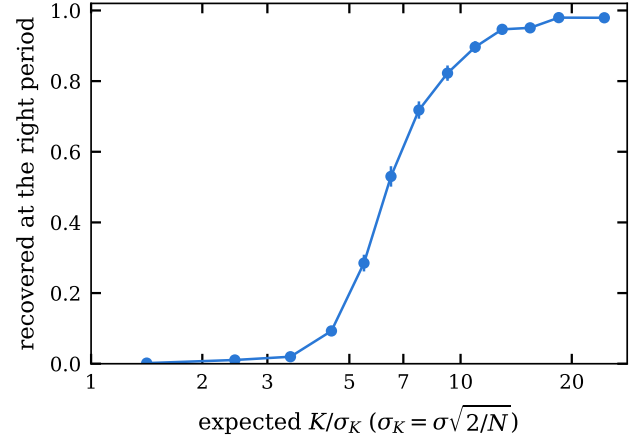


FIG. 2.— Recovery of injected planets (alone) at the right period with FAP below 1%, against the expected K/σ_K .

from the class mix and the amplitude priors, not from the archive. Because both planet and activity amplitudes scale with the star’s scatter, the planet fraction hardly depends on N (56% in the lowest quartile, 58% in the highest), but it would change with the mix. Less dependent on the mix is where activity impostors fall. An evolving spot pattern has lines of width $\sim 1/(\pi\lambda)$, and of the 1736 activity impostors 36% lie inside those lines but more than one resolution element from any exact harmonic, about as many as lie on a harmonic (440 of them excluding the low-frequency envelope). The balance depends on the spot lifetime, and so on its prior: for λ shorter than three rotations, 164 impostors lie inside a line at $k = 1$ –4 and 72 on a harmonic; for λ longer than ten rotations, 120 and 403. A harmonic-distance test does not look inside the lines.

5.4. Calibration of the rungs on planets

For the 1904 planets alone detected at the right period, the block phase probability has a median of 0.56 and falls below 0.05 in 4.4% of cases, close to nominal. With $n_b - 2$ degrees of freedom for the phase, our first choice on the argument that the fitted frequency absorbs a linear phase drift, 8.7% of the same planets fall below 0.05, so we adopted $n_b - 1$. The block amplitude probability is somewhat liberal: its median is 0.44 and 7.4% (6.2–8.6%) fall below 0.05, probably because an eccentric orbit fitted with a sinusoid, and heavy-tailed noise, add scatter the block uncertainties do not include. The half-phase difference exceeds 1.96σ in 1.6% of planets and the half-amplitude difference in 6.2%. Taken alone, the rungs of the a-priori rule reject these planets at the following rates: period 0.4%, phase 1.2%, indicators 1.6% and rotation 0.0%. The indicator rung, nominally a 10^{-3} test, rejects 1.6% because it takes the maximum over indicators and frequencies.

5.5. Which tests carry the information

Table 2 gives the power of each feature alone to separate planets from impostors among significant peaks, as the area under the ROC curve (AUC). No single test is decisive. The strongest, the indicator power (0.78), the block phase rms (0.77) and the growth of $\Delta\chi^2$ (0.76), are only slightly better than the detection statistics themselves (FAP 0.74, $\Delta\chi^2$ 0.75). The distance to a rotation harmonic is informative where it can be computed: for the 679 peaks with a significant rotation proxy it reaches 0.90, but over all peaks only 0.60, because the proxy is usually not detected: it is significant for 28%

TABLE 2
POWER OF EACH LADDER FEATURE ALONE TO SEPARATE PLANETS FROM IMPOSTORS AMONG 4931 SIGNIFICANT SIMULATED PEAKS.

Feature	AUC	Feature	AUC
ind_max_dchi2	0.78	block_phase_p	0.63
block_phase_rms	0.77	apod_gain	0.62
growth_rho	0.76	act_log10_fap	0.62
snr_K	0.76	half_amp_z	0.62
dchi2	0.75	log_ncyc [†]	0.61
log10_fap	0.74	harm_dist	0.60
ind_max_absr	0.71	harm_alias_dist	0.60
alias_margin	0.71	harm_missing	0.59
K_over_rms	0.69	half_phase_z	0.59
half_imbalance	0.68	log_n [†]	0.51
block_amp_p	0.67		

AUC is reported as $\max(A, 1 - A)$, so it does not depend on the sign of the feature, and after the transforms of Sect. 5.6. [†]Reported but not used by the classifiers, because its power depends on the injection priors.

of the activity impostors and 0.6% of the planets alone. The number of epochs alone has an AUC of 0.51: with both planet and activity amplitudes scaled to the star’s scatter, it carries no information about the class.

5.6. Classifiers

We trained a logistic regression (L_2 penalty, $C = 0.5$, standardised features, median imputation with missing-value indicators) and gradient-boosted trees (scikit-learn’s histogram implementation with 400 iterations, learning rate 0.04, 15 leaves, $L_2 = 1$ and at least 40 cases per leaf; Pedregosa et al. 2011) on the ladder features of the significant peaks, excluding $\log N$ and $\log(Tf_0)$. Heavy-tailed features enter as $\log(1 + x)$, the block probabilities as $\log_{10} p$, the alias margin as $\text{sign}(x)\log(1 + |x|)$, and a missing `harm_dist` as $\log(1 + 1000)$ with the flag `harm_missing`. We used five-fold cross-validation in which all cases of a star are in the same fold. Out of fold, the AUC is 0.913 for the logistic regression and 0.917 for the trees (95% intervals 0.903–0.923 and 0.907–0.926 from a bootstrap over stars, which holds the fitted models fixed), against 0.76 for the detection features alone (FAP, $\Delta\chi^2$, K/σ_K) and 0.74 for the FAP alone (Fig. 3). The combination, not any one test, carries the information. Adding $\log N$ and $\log(Tf_0)$ raises the trees’ AUC to 0.926, through the period distributions assumed for planets and rotation. Removing every activity-indicator and rotation feature lowers it to 0.85, and removing only the rotation-proxy features to 0.916; the indicator power dominates the permutation importance (Fig. 5).

The Brier scores are 0.115 and 0.114. Calibration was tested per bin of predicted probability (Fig. 4): for the logistic regression the largest deviation is 1.6σ , with 0 of 10 bins beyond 2σ , so it is well calibrated; the trees are overconfident at both ends, with a largest deviation of 4.2σ and 7 of 10 bins beyond 2σ . Either calibration holds only for the class mix of the simulations, so on real candidates the output is a score, not a probability. At a threshold of 0.5 the trees accept 87% of planets and 22% of impostors: 20% at a rotation harmonic, 13% inside a rotation line, 24% of other activity, 44% of planet aliases and 38% of noise peaks. At a false-positive rate of 5% they accept 61% of planets, and at 1%, 32%. If one significant signal in ten were a planet, the precision at the 0.5 threshold would be 31%; at one in two, 80%.

5.7. The a-priori rule

The rule accepts 96% of planets and 47% of impostors; at the same planet acceptance the trees accept 38% of impostors

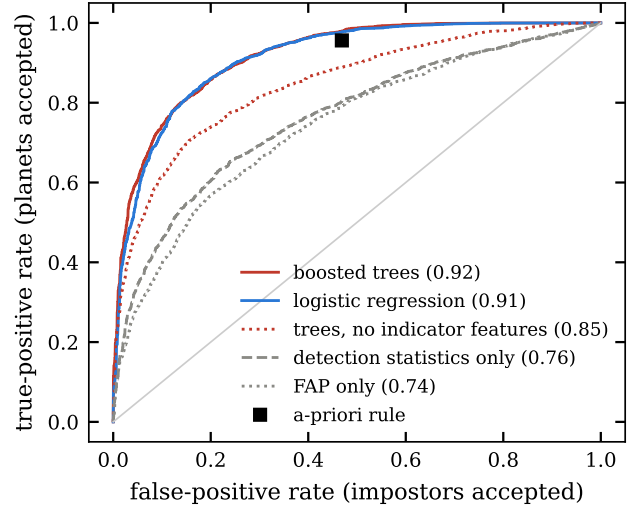


Fig. 3.— Separating planets from impostors among significant simulated peaks, out of fold with folds grouped by star; AUC in brackets. The square is the a-priori rule.

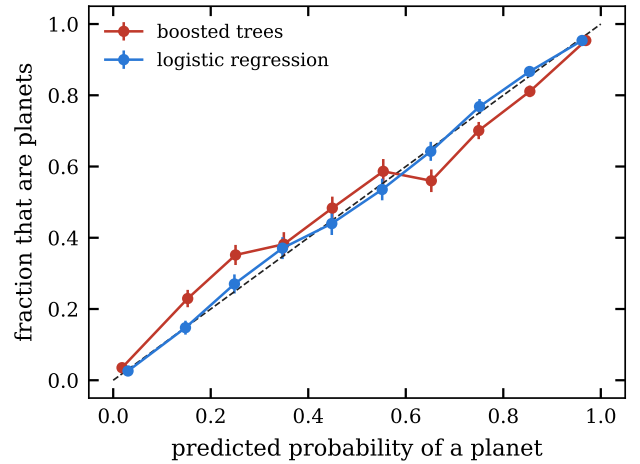


Fig. 4.— Reliability of the out-of-fold probabilities: the fraction of cases that are planets in bins of predicted probability, with binomial errors.

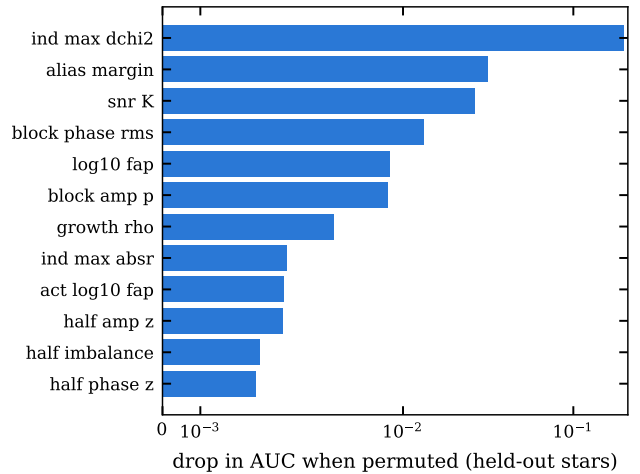


Fig. 5.— Permutation importance of the features for the boosted trees, measured on a quarter of the stars held out of training (symmetric-log scale).

(Fig. 3). It accepts 34% of impostors at a rotation harmonic, 37% of those inside a rotation line, 57% of other activity, 68% of planet aliases and 93% of noise peaks. Its thresholds sit where they reject few planets, and about half of the impostors in these simulations cross none of them: the indicators are weak or silent in many activity cases (Sect. 5.8), a rotation proxy is usually not detected, and a noise peak, once significant, looks stationary.

5.8. Sensitivity to the simulated activity

How much the indicators help depends on how strongly activity shows in them, which the simulations set by assumption, and it is the largest dependence we find. For the 360 activity impostors whose indicators received nothing, the rule accepts 90% and the trees 44%; for indicator amplitudes of 0.1–0.3 times the scatter (446 impostors), 58% and 24%; for 1–3 times (462), 8% and 3%. The RV amplitude of the activity matters too: for amplitudes of 0.2–0.5 times the star’s scatter (74 impostors) the rule accepts 74% and the trees 38%, above once the scatter (1428) 39% and 16%, presumably because weak activity becomes significant mainly where it is coherent. The rotation domain matters as well. For activity impostors with $P_{\text{rot}} > 40$ d (830), the rule accepts 41% and the trees 21%; with $P_{\text{rot}} < 10$ d (276), 44% and 13%. For spot patterns that live longer than ten rotations (686), 37% and 19%; shorter than three (471), 45% and 17%. The 321 impostors with both a long rotation period and long-lived spots, the regime of GJ 581 d, are accepted at 35% by the rule and 22% by the trees, against 41% and 18% for all activity impostors: over the ranges simulated, the rotation period and the spot lifetime matter much less than the indicators.

6. PUBLISHED SIGNALS WITH KNOWN FATES

6.1. Construction

We compiled 46 periodic signals once claimed as planets around HARPS-RVBank stars with at least 20 nights, and labelled each by the later peer-reviewed verdict (Table A1): 19 upheld by independent evidence (transits, another spectrograph or an independent data set), 7 accepted from HARPS data and not contested, 17 refuted (attributed to activity, an alias or the sampling, or not recovered with better data) and 3 disputed, which we exclude from the metrics. Of these signals, 6 were not discovered with HARPS data. The labels, periods and references were compiled with web searches by an AI agent and each was checked against the publisher’s record; we did not choose signals by how the ladder treats them. For every signal we remove the other accepted planets of its system with Keplerian fits and evaluate the ladder at the claimed period.

Three caveats apply. Several refuted signals were refuted with the same kinds of tests the ladder uses, so agreement is partly circular. The refuted signals were claimed from subsets of the data now in the archive, so the benchmark tests the ladder on the full archive, not on the data the discoverers had. And the numbers are small, so we give exact binomial intervals.

6.2. Significance

Whether a claimed signal is significant depends on the question. A blind search of the full archive needs the global FAP over the search band: by that measure 11 of the 17 refuted signals are not significant ($\text{FAP} > 1\%$), while 21 of the 26 upheld signals are (Fig. 6). The 5 upheld signals that are

not have measured K/σ_K of 2.0–5.1 in these data, all were confirmed with other instruments or by transits, and all lie below the 50% completeness point of Fig. 2, although that point is in expected K/σ_K , about which the measured value scatters by roughly one. For a period known in advance, the single-frequency probability is the right measure; below 1% by that measure, 4 refuted and 1 upheld signals are not significant. This probability is taken at the best point within $\pm 0.5/T$ and from the likelihood ratio, which Sect. 5.1 found liberal, so it overstates significance somewhat; Table 4 gives the counts at the claimed period itself. Even so, most refuted signals leave some power at their period in the full archive; what they lack is the strength to stand out in a blind search.

6.3. The rule and the classifiers

Among significant signals, the a-priori rule accepts 1 of the 6 refuted ones (95% interval 0–64%) and rejects 2 of the 21 upheld ones (1–30%): 1 of 14 upheld by independent evidence and 1 of 7 accepted from HARPS alone. Table 3 lists the features of the significant refuted signals and of any significant upheld signal the rule rejects. All 21 significant upheld signals are the highest peak in the band, against 3 of the 6 significant refuted ones, a difference the classifiers, trained on highest peaks, do not see. GJ 581 d, Barnard’s star b at 233 d and HD 41248 c fail on indicator power (the first was attributed to activity by Robertson et al. 2014, the second to a yearly alias of rotation by Lubin et al. 2021, the third to rotation by Santos et al. 2014, whose rotation period our proxy recovers); Kapteyn c fails on phase and has $H\alpha$ power as well; AD Leo b fails the period rung, because a stronger peak of the same activity signal lies a few resolution elements away, and has chromatic-index power, as expected for the spot-induced signal of Carleo et al. (2020). The rule accepts HD 26965 b, for which the indicators show neither power above the threshold at the period nor a significant rotation proxy. It rejects two upheld signals, HD 192310 b and GJ 163 c, on indicator power. Several margins are thin (Table 3). Adding the rotation-alias test of the first benchmark pass (`harm_aliases_dist`) to the rule changes none of these outcomes.

The classifiers rank the significant signals with an AUC of 0.98 (trees; 95% interval 0.93–1.00 from a bootstrap within each class) and 0.99 (logistic regression; 0.95–1.00), misordering 2 and 1 of the 126 upheld–refuted pairs. At a score of 0.5 the trees accept 1 significant refuted signal (GJ 581 d) and reject 2 upheld ones (HD 15337 c and HD 192310 b); the logistic regression accepts 1 (GJ 581 d) and rejects 1 (HD 15337 c, the weakest upheld signal that is significant). GJ 581 d has indicator power above the rule’s threshold but a strong, stationary, alias-free peak, and both models weigh the latter more. Without the 4 upheld signals with $K/\sigma_K > 20$ the trees’ AUC is 0.98; for signals upheld by independent evidence only, 0.98. The model without indicator features reaches 0.82 (0.60–0.98). Among the same significant signals, single features reach AUCs of 0.82 (growth of $\Delta\chi^2$), 0.75 (alias margin), 0.63 (block phase rms), 0.98 (indicator power) and 0.74 (FAP). On signals that are not significant, the classifiers, trained only on significant peaks, are outside their training domain and their scores should not be used.

Only 1 refuted host, AD Leo, has a published rotation period outside the simulated 3–150 d; at 2.23 d it is also below the 2.5-d lower limit of the rotation proxy, so no rotation test is possible for it.

The rotation proxy behaves differently on real stars. It is

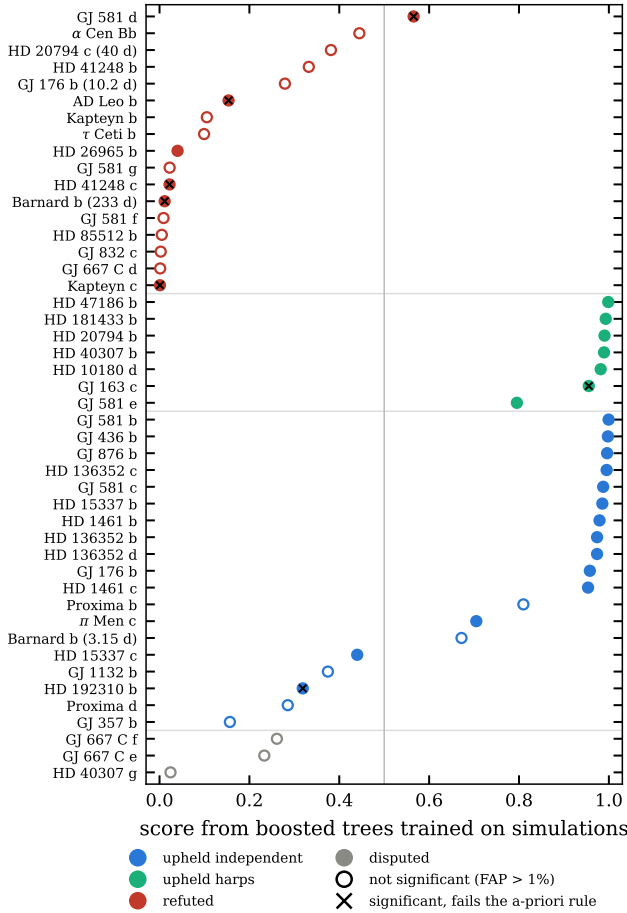


FIG. 6.— The benchmark: score from the boosted trees for signals upheld by independent evidence (blue), accepted from HARPS data (green), refuted (red) and disputed (grey). Open symbols are not significant at the claimed period in a blind search of the full archive; crosses mark significant signals rejected by the a-priori rule.

significant for 18 of the 28 benchmark hosts and for 11 of the 21 significant upheld signals, but for only 0.6% of simulated planets: permutation destroys the star’s own coherent indicator variations, rotation and cycles alike, so simulated planets sit on indicators without activity and real ones do not. Where a host has a significant proxy and a published rotation period, the two agree within 20% for 5 of 10. The rotation features of real candidates therefore lie partly outside the simulated range. A model trained without the rotation-proxy features reaches an AUC of 0.98 (0.93–1.00) on the significant benchmark signals.

6.4. Changes after the first pass

The rule was frozen before the first benchmark pass, and its thresholds have not changed. The pipeline has, four times. After the first pass exposed failures we lowered the S/N cut from 20 to 10 (it removed 15% of the spectra of Proxima Centauri), replaced two-term Fourier prewhitening of known companions by Keplerian orbits (the Fourier terms failed for π Men b with $e = 0.64$), added the quadratic trend and the 15- σ clip (for the binaries α Cen B and GJ 667 C), and added `harm.alias.dist`. An independent review of the first draft then found four errors, which version 0.2 corrected: nights began at 0 h UT instead of local noon; the period rung compared candidates with aliases outside the search band and could re-

fine onto the candidate’s own peak, so that it rejected planets on their one-day mirrors; the jitter was held at its planet-free value; and the block tests had the wrong degrees of freedom. At the same time we widened the window around a claimed period from $\pm 0.25/T$ to $\pm 0.5/T$, after the review showed a refuted signal refined to the edge of the narrower window, and changed the simulations (Sect. 4). A second review found that the effective window had grown to $\pm 0.6/T$, now capped at $\pm 0.5/T$, and that the rotation proxy fired on slow variations of real indicators; the indicators are now detrended like the velocities. A third review found that this detrending, done by least squares, was tilted by indicator outliers in the simulations, and that many real proxies sat at half a year; outliers are now clipped before the fit and the seasonal aliases are excluded from the proxy search (version 0.3.1). Table 4 shows how the benchmark results depend on the window: with the first-pass window the rule rejects 3 of 5 significant refuted signals and 3 of 21 upheld ones; at the claimed period itself, 3 of 4 and 5 of 19. Of the benchmark signals, 6 reach the edge of the adopted window (marked in Table A1).

Table B1 lists the 7 benchmark signals whose outcome differs between the first pass and this version. The frozen-pipeline results are released with the code.

6.5. A test forward in time

The benchmark asks how the ladder treats signals whose fate was already known. A stricter test asks whether it would have pointed the right way before the fate was known. HARPS-RVBank holds spectra taken before 2022 (the last on 2021 December 29), so we ran the frozen ladder blind on every star with at least 20 nights over 100 d (1198 stars; the highest peak per star, 1.2–500 d) and built the list it would have produced at the start of 2022: significant peaks that pass the a-priori rule, have $K < 100 \text{ m s}^{-1}$ and lie on stars with no planet known before 2022. We count a planet as known before 2022 if the NASA Exoplanet Archive dates its discovery paper before 2022 (queried on 2026 October 4) or exoplanet.eu dates its discovery before 2022 (queried on 2026 October 6). The second catalogue also counts earlier announcements, such as planets announced in 2011 (Mayor et al. 2011) that the NASA archive lists only from a later paper or not at all. That list has 102 signals. Since then, the NASA archive has listed 10 of them as planets published at the same period ($|1/P - 1/P_{\text{pub}}| < 1.5/T$).

Ranked by the boosted trees, 6 of the top 10 and 7 of the top 20 were later published, against 1.0 and 2.0 expected from the base rate of 10% (hypergeometric $p = 2.8 \times 10^{-5}$ and 3.4×10^{-4} ; Fig. 7). Ranked by the false-alarm probability alone the counts are 3 and 7, and by the logistic regression 5 and 6. Over the whole list the area under the ROC curve is 0.77 (95% interval 0.53–0.98) for the trees, 0.77 (0.54–0.95) for the logistic regression, 0.81 (0.61–0.94) for the false-alarm probability and 0.79 for K/σ_K . The difference between the trees and the false-alarm probability is -0.03 (paired bootstrap, 95% interval -0.19 to 0.08). The later planets gather at the top of the list, but with 10 positives this test does not show that the classifiers rank better than significance alone.

The concentration at the top survives the other choices we tried. Keeping stars that had other planets before 2022 (126 signals, the same 10 later planets) leaves the trees at 6 of the top 10. Dating planets by the NASA archive alone keeps 13 more stars, among them HD 134606 and HD 137496, whose planets were announced in 2011 and 2021, and the hosts of two TESS planets that exoplanet.eu dates to 2019; it gives 14

TABLE 3

SIGNIFICANT REFUTED SIGNALS, AND SIGNIFICANT UPHOLD SIGNALS THE A-PRIORI RULE REJECTS: THE FEATURES BEHIND EACH VERDICT.

Signal	Verdict	$\log_{10}\text{FAP}$	Alias margin	Indicator $\Delta\chi^2$	P_{proxy} (d)	harm_dist	Fails at	Trees	Logistic	No ind.
GJ 581 d	R	-14.9	59.1	19.59 (contrast)	138	1.4	ind.	0.57	0.66	0.95
Kapteyn c	R	-2.7	5.2	22.90 ($\text{H}\alpha$)	112	5.2	phase	0.00	0.00	0.01
HD 41248 c	R	-10.0	34.4	31.67 (ΔLW)	26	0.9	ind.	0.02	0.05	0.87
HD 26965 b	R	-2.7	17.8	12.16 (R'_{HK})	—	—	—	0.04	0.03	0.06
Barnard b (233 d)	R	-4.6	35.8	20.65 (NaD1)	74	41.8	ind.	0.01	0.01	0.04
AD Leo b	R	-33.9	-3.2	64.04 (CRX)	—	—	period	0.15	0.03	0.79
HD 192310 b	UI	-44.4	142.3	19.49 (R'_{HK})	48	34.1	ind.	0.32	0.81	0.69
GJ 163 c	UH	-11.3	66.3	15.72 (ΔLW)	61	41.8	ind.	0.96	0.68	0.86

Alias margin: $\Delta\chi^2(f_0)$ minus the strongest alias inside the band. Fails at: the first rung of the a-priori rule the signal fails (period, phase; ind.: indicators; rot.: rotation; —: passes). Indicator $\Delta\chi^2$: the largest near f_0 , with the indicator. P_{proxy} : the rotation proxy, when significant. The last three columns are classifier scores.

TABLE 4

BENCHMARK RESULTS FOR THREE WINDOWS AROUND THE CLAIMED PERIOD.

Window	Refuted, significant		Upheld, significant		AUC (significant)		Not significant, single f	
	N	rule rejects	N	rule rejects	trees	logistic	refuted	upheld
$\pm 0.5/T$ (adopted)	6	5	21	2	0.98	0.99	4	1
$\pm 0.25/T$ (first pass)	5	3	21	3	0.97	0.98	5	1
claimed period	4	3	19	5	0.97	0.96	7	3

Significant: global Baluev FAP < 1%. The last two columns count signals whose single-frequency probability is above 1% (of 17 refuted and 26 upheld).

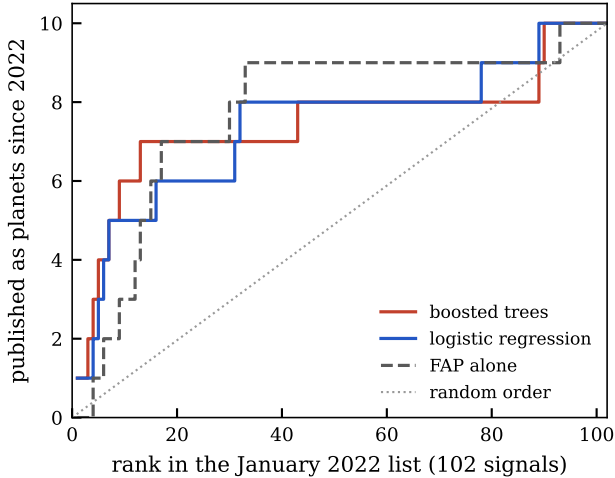


FIG. 7.— The 102 signals the ladder would have listed in January 2022 on stars with no planet known before 2022, ranked by each score: the cumulative number published as planets at the same period since then (10 in all).

later planets among 115 signals, 6 of them in the trees’ top 10. A window of $1/T$ instead of $1.5/T$ changes nothing. On short baselines the window is wide, though: 4 of the later planets differ from the archival period by 11–27%. Requiring the periods to agree within 5% leaves 6 later planets, 4 of them in the trees’ top 10 (0.6 expected, $p = 6.7 \times 10^{-4}$) and 2 in the top 10 by the false-alarm probability.

Two limits apply. Of the 10 later planets, 8 orbit stars with TESS candidates, often observed with HARPS because of the candidate, so the list is enriched in stars someone was already following; 5 of the 6 in the trees’ top 10 are such stars, and without them 2 later planets remain, too few to test. And “not yet published” is not “false”: the share of later planets is a lower bound that may rise as the other signals are followed up.

The open signals also make predictions that new data can test. We froze them, with circular ephemerides, in a list (the discovery ledger) and tested the two of its 30 highest-ranked signals that had at least ten public spectra taken after 2022

in the ESO archive, with the decision rule written down before their velocities were read. For GJ 902 (36.10 d, predicted $K = 1.90 \pm 0.41 \text{ m s}^{-1}$), ESPRESSO, HARPS and NIRPS spectra on 30 nights (41 nightly velocities, one per instrument and night) give $0.20 \pm 0.37 \text{ m s}^{-1}$ at the predicted phase. One ESPRESSO product in the search cone belongs to another star (-106 against $+70 \text{ km s}^{-1}$) and was left out, a decision taken after the headers were read. The ladder and the rule passed this signal (trees score 0.86), and the new data exclude it. For HD 58489 (19.51 d, $5.5 \pm 1.3 \text{ m s}^{-1}$), 18 HARPS nights give $4.0 \pm 1.0 \text{ m s}^{-1}$ at the predicted phase, about 2.5σ once the statistic is calibrated on noise-only simulations. That result gives the two cross-correlation masks of these spectra separate zero points, a choice made after the velocities were displayed but before any fit. With the single zero point that was registered, the amplitude is $1.0 \pm 2.9 \text{ m s}^{-1}$ and the test is undecided, which is the registered verdict. Even with separate zero points, a sinusoid at a random phase does as well in 17% of trials, and 13% of periods between 2 and 150 d fit the new velocities better. One prediction failed and the other is undecided. The ledger, the rule, the data and the results are released with the code; git commit times record the order in which they were written, but they are not independent timestamps.

7. A WORKED EXAMPLE: HD 297396 B

HD 297396 b is a 4.27-d candidate found in a periodogram survey of HARPS-RVBank (Fraser, in preparation). Through the generic pipeline, with 106 nightly epochs over 17.9 yr (the discovery analysis, with stricter cuts, keeps 104), its peak has a Baluev FAP of $10^{-5.7}$ and $K = 6.6 \pm 0.9 \text{ m s}^{-1}$ ($K/\sigma_K = 7.6$). It beats its aliases by $\Delta\chi^2 = 10.6$, holds phase across halves (0.3σ) and blocks ($p = 0.96$), grows steadily with the number of epochs ($\rho = 0.97$), gains $\Delta\chi^2 = 0.4$ from an envelope, and no indicator has power near its period beyond $\Delta\chi^2 = 5.6$. The indicators give no significant rotation proxy, so the rotation rung is not applied. It passes the a-priori rule, whose thresholds were partly set during its own analysis, and the classifiers give it scores of 0.99 (trees), 0.88 (logistic) and 0.96 (without indicators). These are not the candidate’s adopted values. The discovery analysis (Fraser, in preparation) gives each of

four groups of programmes its own offset and jitter and finds $K = 5.5 \pm 0.8 \text{ m s}^{-1}$. Its false-alarm probability, calibrated by simulation, is 1.4×10^{-3} with all epochs; without one discrepant night no simulation exceeds the peak, and the analytic bound is 7×10^{-6} . Most of the gap comes from the jitter. The generic statistic refits the jitter scale with the signal in the model; with the jitters held at their planet-free values, the same pipeline gives $10^{-3.8}$ with all epochs and $10^{-5.9}$ without the discrepant night (Sect. 5.1 found the first treatment liberal and the second conservative on permuted real noise). We do not break down the remaining factor of 5–15: the two analyses also differ in their noise models (two eras against four groups of programmes), their epochs and their frequency bands.

The example also shows what the ladder cannot do. The star’s rotation period is not measured; the indicators are weak for this star; and activity calibrations place its rotation near 39 d, close to nine times the orbital period, a harmonic that our simulated activity rarely puts power into (Fraser, in preparation). A high score says that the signal behaves like the simulated planets and unlike the simulated impostors; it does not replace a second instrument.

8. SOFTWARE AND VERIFICATION

8.1. The package

The ladder, the simulations, the rule and the classifiers form a Python package, `rvvet` (version 0.3.1), with a command-line interface that writes the features of one star as JSON together with a provenance block (package version, a hash of the source files, the input file and the arguments). It has 40 unit tests. Thirteen compare the code with an expected value obtained another way: the projected periodogram against dense least squares, its invariance to era offsets, the nightly means, a circular Keplerian against a sinusoid, the analytic (G, G') covariance against finite differences, the single-frequency limit of the FAP, the ordering of the likelihood-ratio and fixed-jitter statistics, the mean of the homogeneity statistic against its χ^2 expectation, and hand-worked cases for the night boundary, the harmonic distances, the rule and the labels. The other 27 check behaviour on constructed inputs: recovery of injected periods, amplitudes and jitters, the Baluev approximation not exceeding its nominal rate for Gaussian noise of known variance, the alias rung staying inside the search band, the window around a claimed period, the detrending and clipping of indicators, the exclusion of seasonal aliases from the rotation proxy, a transient versus a stationary signal, removal of an $e = 0.6$ orbit, deterministic seeding, folds that never split a star, and the command line. A simulated case takes about 0.6 s on one core. The results come from Python 3.13 with `numpy` 2.4, `scipy` 1.17, `pandas` 3.0, `scikit-learn` 1.8 and `matplotlib` 3.10, pinned in the release; in that environment the released code regenerates the benchmark features, 20 randomly chosen simulated cases and every number, table and figure exactly. Newer versions differ slightly: with `scikit-learn` 1.9.1, which assigns stars to cross-validation folds differently and builds different trees, the classifier AUCs on the simulations move by at most 0.003 and the benchmark AUC without indicator features changes from 0.82 to 0.78. Every computed result in the text of this paper is a macro written by the analysis scripts from their outputs (Sect. 6.5 by its own script), and so is every table body except Table 1; the numbers typed by hand are design constants, literature values and counts of items such as tests, listed in the released provenance file.

8.2. Defects found in an AI-assisted analysis

The HD 297396 b analysis that motivated this work was carried out by one author with an AI assistant (Claude, Anthropic) that wrote most of the analysis code and draft text, while the author made the analysis decisions. Its revisions were checked in up to four ways: *self-review*, in which the changed numbers were checked against their scripts, source papers and catalogues; an *independent reviewer*, a separate instance of the model given the manuscript, the source papers and the scripts but not the conversation that produced them, and asked to act as a referee; an *external report*, a review of version 13 (the “Referee-Lens” report) whose requests version 14 implemented; and, for version 15, an automated *reference check* of every entry against the publisher’s record. Each version has a change log.

Method. We extracted the defects from the records written at the time: the change logs of versions 13, 14 and 15, the referee report on version 15 and the reference check. A codebook fixed before any coding (released with the code) defines the unit: a statement, number, figure element, reference or analysis step that a record says was changed because it was wrong, unsupported, stale, inconsistent or ambiguous, counted once per root cause. New analyses, restructuring and cosmetic fixes are not defects. An instance of the model that had not seen any earlier tabulation extracted 84 defects, listing the 99 record items it considered and excluded, with reasons; who found each defect (the *route*) and whether it was in a released version or caught in a draft (the *stage*) are read from the records. Two further instances, blind to each other, coded each defect’s category and origin; they are instances of the same model, so their agreement measures the consistency of the codebook’s application rather than independent human judgement. They agreed on 96% of the categories (Cohen’s $\kappa = 0.95$) and 94% of the origins ($\kappa = 0.92$); the 6 defects with any disagreement were settled by a third reading, with the reason recorded. One coder read two rows of the list of excluded items before coding; neither row is a coded defect.

Results. Table 5 gives the 84 defects by category and route; 48 were in released versions and 36 were caught in drafts. The largest category, 46% of the total, is overclaims (39), statements stronger than the evidence or caveats that were missing: a simulation presented as testing a scenario its kernel could not produce, a weak out-of-sample test described as agreement, companions said to be excluded out to 100 au when the contrast curve stopped near 55 au. Revising the text was itself a source: 21 defects were introduced by revision, 14 of them overclaims, mostly caveats that the rewrite of version 14 dropped. Of the 19 numerical errors (transcription and arithmetic), 8 came from the literature, such as parameters of comparison planets that did not match their source papers, 5 from calculations in the analysis, 2 from revision, and 4 were numbers the project’s own code had produced correctly and that were carried into the text wrongly, among them the candidate’s own semi-amplitude pasted into the column of a comparison planet. Over all categories, only 5 defects had that origin; errors of method, such as an activity index on one calibration scale fed into a relation built on another, came from the analysis itself.

The independent reviewer found 53 of the 84, including 22 of the 39 overclaims; self-review found 13, the external report 3 and the reference check 2; for 13 the record does not say. These shares depend on the order of the checks, since the reviewer saw what self-review had not caught, and on which

TABLE 5
DEFECTS LOGGED IN THREE REVISIONS OF AN AI-ASSISTED ANALYSIS, BY CATEGORY
AND BY THE CHECK THAT FOUND THEM.

Category	Self	Reviewer	Report	Refs	Unattr.	Total
Overclaim	2	22	3	0	12	39
Transcription	5	7	0	0	0	12
Method	3	6	0	0	1	10
Arithmetic	2	5	0	0	0	7
Wording	1	6	0	0	0	7
Stale	0	4	0	0	0	4
Reference	0	1	0	2	0	3
Reasoning	0	2	0	0	0	2
Total	13	53	3	2	13	84

From the change logs of versions 13–15 of the HD 297396 b manuscript, the referee report on version 15 and its reference check (released with the code). “Reviewer”: an independent instance of the model acting as a referee; “Report”: an external review of version 13; “Unattr.”: the record does not say who found it.

record exists for each version (for version 15 the main record is the referee report); the records were written by the same assistant whose work they describe. The log counts the defects that were found, not those that remain, for one project and one model.

The same checks were applied to this paper. The first draft reported 30 defects, not 84: it had condensed the records by hand, its route column contradicted its own notes, and it concluded that most defects arose where numbers passed from code into text. That draft also contained four errors in the pipeline (the night boundary, the alias rung, the fixed jitter and the degrees of freedom of the block test; Sect. 6.4). The independent review of the draft found all of them, and two further rounds of review found more (Sect. 6.4). The night-boundary error also affected two archive-wide numbers in the analysis of HD 297396 b, though no result about the star; it was corrected there after release. The log describes one project rather than testing a hypothesis, but it is consistent with two practices: a reviewer that sees the sources but not the history, which found most of what was found, including most overclaims; and checks aimed at what the text claims, not only at whether its numbers match the code, since generating every number from the analysis outputs, as this paper now does, addresses only a small share of the defects.

9. LIMITATIONS

The simulated activity. The quasi-periodic FF’ model is a common description of spot-driven variability, but it is an assumption, and so is every prior on its parameters. Most consequential is how activity imprints on the indicators: we vary the amplitude, the lag and the share of indicator-free cases (Sect. 5.8), but the fifth of cases without any indicator signal is an arbitrary choice, and the classifiers learn the coupling we assumed. The kernel also puts little power in high harmonics of the rotation period.

The noise model. Permuting residuals within eras keeps the error distribution, its tails and the joint RV–indicator noise, but removes correlated noise on timescales longer than a night, including magnetic cycles and slow instrumental drifts. The false-alarm rates of Sect. 5.1 are for noise without such correlations.

The priors. The fraction of significant peaks that are imposers, and the classifiers’ probabilities, depend on the class mix and the amplitude priors. They describe these simulations, not the archive.

The benchmark. It has 43 signals with a verdict and only 6 significant refuted ones, and its results depend on the window

allowed around a claimed period (Table 4). The rotation proxy behaves differently on real indicators than on simulated ones (Sect. 6). Its labels come from later literature that often used similar tests, and some of its upheld signals are easy, 4 with $K/\sigma_K > 20$. It measures agreement with the literature on the full archive, not the performance a discoverer would have had.

Changes after the first pass. The rule’s thresholds were fixed before the benchmark was run, but the pipeline changed four times afterwards (Sect. 6.4), and some thresholds came from the analysis of HD 297396 b.

One instrument. Every test here uses HARPS data alone. A signal specific to one star in one spectrograph passes every rung; only a second instrument addresses it.

10. CONCLUSIONS

1. Neither common treatment of the jitter gives nominal false-alarm rates on real archival noise. Held at its planet-free value, it makes the Baluev approximation conservative by a factor of 4–8; rescaled under the signal model, liberal by a factor of 1.8 at 1% and 3.3 at 0.1%. On these stars a Baluev FAP of 0.44% corresponds to a real false-alarm rate of 1%.
2. Evolving activity produces many significant peaks inside the broad lines of its power spectrum rather than on exact rotation harmonics; the balance shifts from lines to harmonics as spots live longer (164 against 72 for lifetimes under three rotations, 120 against 403 above ten). Tests of harmonic distance see only the harmonics, and only when the indicators reveal the rotation period.
3. No single test is decisive: the best reaches an AUC of 0.78 and the detection statistics 0.74–0.75. Combined, the rungs reach 0.913–0.917, and 0.85 without the indicators. How strongly activity shows in the indicators matters more than any other property of the simulated activity.
4. On published signals, 11 of 17 refuted signals are not significant in a blind search of the full archive, though 13 keep single-frequency power at their period. The rule with thresholds fixed in advance rejects 3 of the 6 significant refuted signals on indicators, 1 on period and 1 on phase, accepts 1, and rejects 2 of 21 significant upheld signals; the classifiers rank the significant signals with AUCs of 0.98 (trees) and 0.99 (logistic). These counts shift with the window allowed around the claimed period, and the numbers are too small to bound either error rate tightly.
5. Run on the archive as it stood when it closed, the ladder’s list held 10 signals later published as planets, gathered at the top (6 of the ten ranked highest by the trees, against 1.0 expected), but the classifiers did not rank them better than the false-alarm probability alone, and 5 of those 6 orbit stars with TESS candidates that observers were already following. Of two open signals tested with newer spectra under a rule written in advance, one was refuted and the other is undecided under the registered model (about 2.5σ with a zero-point split chosen after the velocities were displayed).
6. In the AI-assisted analysis that motivated this work, 84 defects were found in three revisions. Overclaims

were the largest category (39), and transcriptions of the code’s own output were few (5); an independent reviewer that saw the sources but not the history found 53 of them.

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Use of generative AI. The author designed the study and made the analysis decisions. The code, the literature search for the benchmark, the extraction and coding of the defect log, and the draft text were produced with the assistance of Claude (Anthropic), a large language model. They were validated as follows. The package has 40 unit tests, 13 of which compare it with an independent calculation (Sect. 8). Every simulated case is reproducible from its seed: 20 randomly chosen cases

regenerated with the released code gave features identical to the stored ones. Every benchmark label, period and reference, and every entry of the bibliography, was checked against the publisher’s record. A separate instance of the model, given the manuscript, the code and the outputs but not the conversation that produced them, reviewed the first draft as a referee; its report and our response are released with the code. The defect log was extracted and coded by separate instances following a codebook fixed in advance, with agreement measured between two coders. Every computed result in the text is generated by the analysis scripts from their outputs; design constants, literature values and counts typed by hand are listed in the released provenance file. The author takes responsibility for the content.

DATA AND CODE AVAILABILITY

HARPS-RVBank is available at the CDS (J/A+A/683/A125). The package *rvvet*, the simulation outputs, the false-alarm calibration, the benchmark table with its references, the frozen rule and the frozen-pipeline results, the defect log with its codebook, sources and both codings, the referee report, the forward test of Sect. 6.5 (the archive run, the catalogue cross-matches, the discovery ledger and the prospective test with its velocities), the pinned software environment and every script used for this paper are released at <https://github.com/samsonmaximus/rvvvet> and archived at Zenodo (doi:10.5281/zenodo.23249395). The benchmark of published signals and the January 2022 list of Sect. 6.5 are also ancillary files of the arXiv version.

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APPENDIX

A. THE BENCHMARK

Table A1 lists the 46 benchmark signals, their verdicts and references, and the outcome of the ladder on the full HARPS-RVBank series of each host.

B. BENCHMARK OUTCOMES BEFORE AND AFTER THE CORRECTIONS

Table B1 lists the benchmark signals whose significance or first failed rung changed between the frozen first-pass pipeline (rvvet 0.1) and this version.

TABLE A1

THE BENCHMARK: SIGNALS AROUND HARPS-RVBANK STARS WITH LATER VERDICTS, AND THE LADDER ON THE FULL HARPS-RVBANK SERIES.

Signal	P (d)	K (m s ⁻¹)	Verdict	N	$\Delta f/T$	$\log_{10}$ FAP	K/σ_K	Fails at	Trees	Discovery; verdict
Barnard b (3.15 d)	3.153	0.55	UI	213	-0.36	0.0	3.4	det.	(0.67)	González Hernández et al. 2024; Basant et al. 2025
GJ 1132 b	1.629	2.76	UI	120	+0.18	-0.3	4.1	det.	(0.37)	Berta-Thompson et al. 2015
GJ 176 b	8.784	4.12	UI	106	+0.50 ^e	-5.2	7.4	-	0.96	Forveille et al. 2009; Rosenthal et al. 2021; Trifonov et al. 2018
GJ 357 b	3.931	1.52	UI	53	+0.04	-0.0	3.3	det.	(0.16)	Luque et al. 2019
GJ 436 b	2.644	18.1	UI	135	+0.16	-76.9	45.6	-	1.00	Butler et al. 2004; Gillon et al. 2007
GJ 581 b	5.369	12.3	UI	248	+0.00	-117.4	46.5	-	1.00	Bonfils et al. 2005; Rosenthal et al. 2021; von Stauffenberg et al. 2024
GJ 581 c	12.92	3.1	UI	248	+0.11	-34.0	16.1	-	0.99	Udry et al. 2007; Rosenthal et al. 2021; von Stauffenberg et al. 2024
GJ 876 b	60.85	239	UI	248	-0.23	-59.9	23.8	-	1.00	Marcy et al. 1998; Rosenthal et al. 2021
HD 136352 b	11.58	1.59	UI	217	+0.18	-21.3	12.4	-	0.97	Udry et al. 2019; Kane et al. 2020
HD 136352 c	27.58	2.65	UI	217	+0.01	-34.9	16.8	-	1.00	Udry et al. 2019; Kane et al. 2020
HD 136352 d	107.6	1.35	UI	217	+0.31	-13.9	10.1	-	0.97	Udry et al. 2019; Delrez et al. 2021
HD 1461 b	5.771	2.32	UI	266	-0.13	-28.7	14.4	-	0.98	Rivera et al. 2010; Rosenthal et al. 2021
HD 1461 c	13.51	1.49	UI	266	+0.13	-8.0	7.9	-	0.95	Díaz et al. 2016; Rosenthal et al. 2021
HD 15337 b	4.756	3.08	UI	116	-0.02	-9.2	9.1	-	0.99	Gandolfi et al. 2019; Dumusque et al. 2019
HD 15337 c	17.18	2.16	UI	116	+0.20	-2.1	5.8	-	0.44	Gandolfi et al. 2019; Dumusque et al. 2019
HD 192310 b	74.39	4.07	UI	402	+0.04	-44.4	17.4	ind.	0.32	Howard et al. 2011; Pepe et al. 2011
Proxima b	11.19	1.38	UI	229	-0.34	-1.1	5.1	det.	(0.81)	Anglada-Escudé et al. 2016; Suárez Mascareño et al. 2020, 2025
Proxima d	5.122	0.39	UI	229	-0.50 ^e	0.0	2.0	det.	(0.29)	Faria et al. 2022; Suárez Mascareño et al. 2025
π Men c	6.268	1.58	UI	209	+0.01	-2.9	5.8	-	0.71	Huang et al. 2018; Gandolfi et al. 2018
GJ 163 c	25.63	2.75	UH	201	-0.05	-11.3	9.2	ind.	0.96	Bonfils et al. 2013
GJ 581 e	3.148	1.8	UH	248	-0.24	-9.4	8.4	-	0.80	Mayor et al. 2009a; von Stauffenberg et al. 2024
HD 10180 d	16.36	2.86	UH	269	+0.15	-34.7	16.0	-	0.98	Lovis et al. 2011
HD 181433 b	9.374	2.94	UH	189	-0.10	-30.2	15.9	-	0.99	Bouchy et al. 2009; Horner et al. 2019
HD 20794 b	18.31	0.614	UH	546	-0.28	-5.2	6.7	-	0.99	Pepe et al. 2011; Nari et al. 2025
HD 40307 b	4.311	1.79	UH	260	-0.01	-23.3	12.8	-	0.99	Mayor et al. 2009b; Díaz et al. 2016
HD 47186 b	4.085	9.12	UH	147	-0.01	-42.6	22.3	-	1.00	Bouchy et al. 2009
AD Leo b	2.226	19.1	R	124	+0.40	-33.9	19.9	period	0.15	Tuomi et al. 2018; Carleo et al. 2020
Barnard b (233 d)	232.8	1.2	R	213	+0.46	-4.6	6.7	ind.	0.01	Ribas et al. 2018; Lubin et al. 2021; González Hernández et al. 2024
GJ 176 b (10.2 d)	10.24	11.6	R	106	+0.13	0.0	2.8	det.	(0.28)	Endl et al. 2008; Forveille et al. 2009
GJ 581 d	66.8	2.63	R	248	-0.01	-14.9	10.2	ind.	0.57	Udry et al. 2007; Mayor et al. 2009a; Robertson et al. 2014; von Stauffenberg et al. 2024
GJ 581 f	433.0	1.3	R	248	+0.14	0.0	1.6	det.	(0.01)	Vogt et al. 2010; Robertson et al. 2014; von Stauffenberg et al. 2024
GJ 581 g	36.56	1.29	R	248	+0.50 ^e	-0.0	3.2	det.	(0.02)	Vogt et al. 2010; Robertson et al. 2014; von Stauffenberg et al. 2024
GJ 667 C d	91.61	1.52	R	263	+0.47	-0.7	4.8	det.	(0.00)	Anglada-Escudé et al. 2013; Robertson & Mahadevan 2014
GJ 832 c	35.68	1.79	R	180	-0.16	-0.0	3.7	det.	(0.00)	Wittenmyer et al. 2014; Gorrini et al. 2022
HD 20794 c (40 d)	40.11	0.56	R	546	-0.50 ^e	-0.0	4.0	det.	(0.38)	Pepe et al. 2011; Nari et al. 2025
HD 26965 b	42.38	1.81	R	86	+0.00	-2.7	6.2	-	0.04	Ma et al. 2018; Laliotis et al. 2023; Burrows et al. 2024
HD 41248 b	18.36	2.93	R	163	-0.17	-0.2	4.3	det.	(0.33)	Jenkins et al. 2013; Santos et al. 2014; Faria et al. 2020
HD 41248 c	25.65	1.84	R	163	+0.22	-10.0	9.0	ind.	0.02	Jenkins et al. 2013; Santos et al. 2014; Faria et al. 2020
HD 85512 b	58.43	0.769	R	580	+0.50 ^e	-1.2	4.9	det.	(0.01)	Pepe et al. 2011; Laliotis et al. 2023
Kapteyn b	48.62	2.25	R	217	-0.25	-0.0	4.0	det.	(0.11)	Anglada-Escudé et al. 2014; Robertson et al. 2015; Bortle et al. 2021
Kapteyn c	121.5	2.27	R	217	-0.49	-2.7	5.8	phase	0.00	Anglada-Escudé et al. 2014; Bortle et al. 2021
α Cen Bb	3.236	0.51	R	305	-0.13	0.0	2.0	det.	(0.44)	Dumusque et al. 2012; Rajpaul et al. 2016
τ Ceti b	13.96	0.64	R	507	+0.23	0.0	2.1	det.	(0.10)	Tuomi et al. 2013b; Feng et al. 2017
GJ 667 C e	62.24	0.92	D	263	+0.36	-0.0	4.2	det.	(0.23)	Anglada-Escudé et al. 2013; Feroz & Hobson 2014; Robertson & Mahadevan 2014
GJ 667 C f	39.03	1.08	D	263	-0.02	-0.0	3.9	det.	(0.26)	Anglada-Escudé et al. 2013; Feroz & Hobson 2014; Robertson & Mahadevan 2014
HD 40307 g	197.8	0.95	D	260	-0.50 ^e	0.0	2.6	det.	(0.02)	Tuomi et al. 2013a; Díaz et al. 2016

P , K : as claimed. Verdict: UI, upheld by independent evidence; UH, accepted from HARPS data and not contested; R, refuted; D, disputed. N , $\log_{10}$ FAP (global, over 1.2–500 d; 0.0 means a FAP near 1) and K/σ_K : this work, at the claimed period. $\Delta f/T$: offset of the evaluated frequency from the claimed one, in resolution elements; ^e marks the edge of the $\pm 0.5/T$ window. Fails at: the first rung of the a-priori rule the signal fails (det.: detection; ind.: indicators; rot.: rotation; -: passes). Trees: score from the boosted trees, in brackets where the signal is not significant and the score is outside the training domain. References: discovery; later verdict. A single reference means the discovery paper itself provides the verdict (e.g. transits) or, for UH, that no later paper contests it.

TABLE B1

BENCHMARK SIGNALS WHOSE OUTCOME CHANGED BETWEEN THE FIRST PASS (RVVET 0.1) AND THIS VERSION.

Signal	Verdict	rvvet 0.1		rvvet 0.3.1	
		$\log_{10}$ FAP	Fails at	$\log_{10}$ FAP	Fails at
Barnard b (233 d)	R	-3.4	period	-4.6	ind.
HD 26965 b	R	-1.8	det.	-2.7	-
Kapteyn c	R	-1.2	det.	-2.7	phase
GJ 163 c	UH	-9.0	-	-11.3	ind.
GJ 176 b	UI	-1.3	det.	-5.2	-
HD 15337 c	UI	-1.3	det.	-2.1	-
HD 192310 b	UI	-33.1	-	-44.4	ind.

Fails at: the first rung of the a-priori rule the signal fails (det.: detection; ind.: indicators; rot.: rotation; -: passes).